Development of Operation Guidelines for Leader-Follower Autonomous Truck-Mounted Attenuator Vehicles

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ABSTRACT

The rapid advancements of connected and autonomous vehicle (CAV) technologies, although possibly years away from wide application to the general public travel, are receiving attention from many State Departments of Transportation (DOT) in a niche area of using Autonomous Maintenance Technology (AMT) to reduce fatalities of DOT workers in work zone locations. Although promising results are shown in testing and deployments in multiple States, current Autonomous Truck Mounted Attenuator (ATMA) system operators are not provided with much practical driving guidance on how to drive this new vehicle systems in a way that is safe to both the public and themselves. To this end, this manuscript aims to model and develop a set of rules and instructions for ATMA system operators, particularly when it comes to critical locations where essential decision making is needed. Specifically, three technical requirements are investigated, including those on car-following distance, critical lane-changing gap distance, and intersection clearance time. Newell's simplified car-following model, and the classic lane-changing behavior model are modified, with roll-ahead distance (RAD) taken into account, to model the driving behaviors of the ATMA vehicles at those critical decision-making locations. Data are collected from real-world field testing to calibrate and validate the developed models. The modelling outputs suggest important thresholds for ATMA system operators to follow. For example, on a freeway with a speed limit of 70 mph and ATMA operating speed of 10 mph, car-following distance should be no less than 75 ft for the lead truck and 100 ft for the follower truck, the critical lane-changing gap distance is 912 ft, and a minimum intersection clearance is 15 seconds, which are all much higher than the requirements for a general vehicle.

Keywords: Connected and autonomous Vehicles (CAV), Autonomous Maintenance Technology (AMT), Autonomous Truck Mounted Attenuator (ATMA), Work Zone, Driving Guideline;

1. INTRODUCTION

Frequent crashes in the nation's work zones pose a great threat to the safety of the public as well as roadway maintenance workers. Nationally, a total of 94,000 crashes occurred in the whole nation's work zones in 2017 alone, resulting in 37,000 injuries and 809 fatalities. Many of these crashes involved state Department of Transportation (DOT) workers. For example, in the State of Missouri, slowing-moving operation vehicles have been crashes into more than 80 times since 2013, resulting in many injuries to DOT workers [1]. One fundamental issue for the DOT worker injuries is that, the speed of mobile and slow-moving operation vehicles generally ranges from 5 mph to 15 mph. When compared with the speed of the faster moving general vehicles on a freeway or a highway, dramatic speed differences are observed. As such, drivers of general vehicles are required to take immediate action, when approaching work zones, such as slowing down or changing lanes. If, due to reasons like driving under the influence [2], such action is not taken in a timely manner, crashes in work zones are more likely to happen. Research suggests that aggressive or distracted driving does occur very often and is the primary factor in crashes in work zones. Other reasons, such as the performance of heavy vehicles, speeding, and dynamic traffic conditions may also result in work zone crashes [3-6]. How to reduce hazards and achieve a safer environment for the DOT workers remains an urgent problem.

The rapid advancements of connected and autonomous vehicles (CAV) technologies, although potentially years away from wide application to the general public travel, are receiving attention from many State DOTs in a niche area of using Autonomous Maintenance Technology (AMT) to reduce fatalities of DOT workers in work zones. One specific technology, named Autonomous Truck Mounted Attenuator (ATMA) vehicle system, is being tested and deployed in multiple States. Figure 1 below shows some photos of an ATMA vehicle system under operation on a roadway, in which (a) is an overview of the system, (b) is a view from above, taken by a drone, and (c) is a view from the rear. The vehicle system includes a manned leader truck, an unmanned follower truck, and a truck mounted attenuator (TMA) installed on the follower truck. The leader truck is designed to take on the normal maintenance work, such as roadway striping and pothole fixing, whereas the follower truck is doing autonomous car-following and serving as a buffer in case a crash happens. The driving distance between the follower truck and the leader truck ranges from a few hundred feet to half a mile or so, and the follower truck is designed to mimic the behavior of the leader truck and drive autonomously. The follower truck is capable of not only speed control (e.g., acceleration, deceleration, and maintaining certain speed), in a similar way as adaptive cruise control (ACC), but also lateral control for the vehicle to stay in the middle of a lane. Actuators, software, electronics, and vehicle-to-vehicle (V2V) communication equipment are installed on the leader truck and the follower truck and, together, they enable the connectivity (mainly V2V) and the autonomous driving capabilities in a leader-follower style.

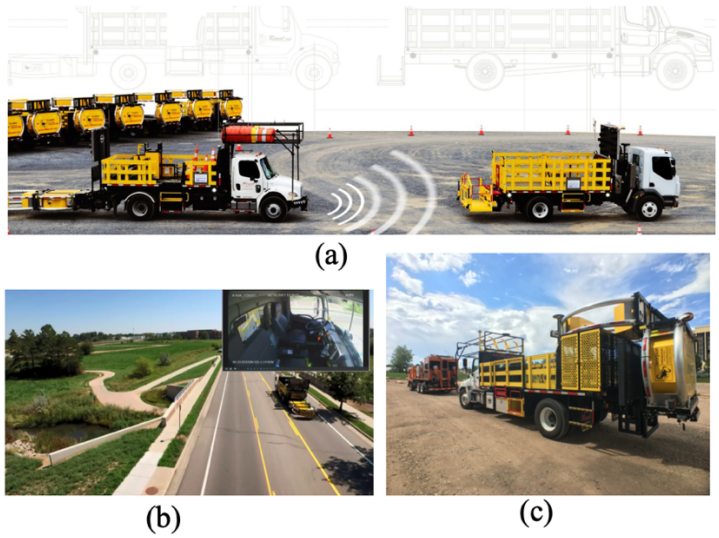


Figure 1. ATMA Vehicle System: (a) system overview; (b) view from above; and (c) view from the rear.

While early research shows that promising results are observed in current testing and deployments in multiple States (for example, the statistical analysis in [7] suggested that the system functioned consistently in a field-testing performed in Sedalia, Missouri,), research on how DOT workers should operate such new system on a roadway network is very limited. The lack of practical operation guidelines may bring considerable safety concerns. This is due to the uniqueness of this system, in that, in essence, the leader-follower system design imposes more requirements on the leader truck drivers, in order to ensure a safe and smooth system operation. The lead truck driver is now required to make driving decisions, not only from his/her own vehicle's perspective, but also to consider the potential implications of these decisions on the follower truck. For example, when vehicles are making turns at intersections or needing to make lane changes, a regular passenger vehicle driver can simply find an acceptable headway gap in the traffic flow and proceed with his/her desired action. However, for an ATMA leader truck driver, he/she needs to avoid a scenario where only the leader truck can proceed, while the follower truck has to remain in its previous location and wait for an acceptable gap to occur. In order words, an ATMA driver will have to wait for a gap larger than normal to make certain that both the leader truck and the follower truck can proceed with a desired action together. Another example is the car-following distance gap requirement. According to classic car-following models, the required minimum car-following gap is dependent on the operating speed and drivers' perception and response time. However, for an ATMA vehicle system, such a minimum gap should not only consider the impact of the follower truck's roll-ahead distance (RAD), but also the accuracy and reliability of the system in maintaining the following distance. Due to the nature of autonomous driving capabilities, the required gap relies less on the human perception and response time.

To bridge this important gap, this manuscript aims to model and develop a set of practical operation guidelines for ATMA system operators, particularly when it comes to critical locations where decision making is needed. Specifically, technical requirements under three scenarios are investigated. We first focus on the car-following distance, i.e., what is a safe minimum distance that a follower truck should keep away from the leader truck, and what is the minimum distance that the leader truck should keep away from the general vehicle in front? Next, we examine the

critical lane-changing gap distance, i.e., what minimum headway gap does an ATMA vehicle system need, before both vehicles can safely proceed with a desired lane changing action? Last, but not least, we investigate when the ATMA vehicle system approaches an intersection, how much time is needed for the ATMA vehicle system to safely cross the intersection? This is important, especially when the vehicles are encountering an ending green light. Or, they may get cut off by vehicles with conflicting movements.

Traffic flow models are developed to work with the unique ATMA vehicle system design. Traditional models, specifically, the Newell's simplified car-following model and the classic lane-changing behavior model, are modified to model the driving behaviors of ATMA vehicles at critical decision-making locations. These changes are necessary, because as discussed above, human driver's perception and response time is no longer a valid aspect to consider, but rather, the accuracy and reliability of the system in maintaining the following distance should be taken into consideration. In addition, the distance between a leader truck and a follower truck also plays an important role in the lane-changing process, as we now need both vehicles to switch to a different lane, before the process can be said to be complete. Finally, RAD, defined as the distance that the follower truck will roll ahead if struck by an errant vehicle, should be correctly modeled and taken into consideration.

This manuscript is organized as follows. First, the ATMA system was briefly reviewed, followed by a review of other relevant literature in Section 2. In Section 3, some preliminaries are presented, including the Newell simplified car-following model, the critical gap for lane changing, and roll-ahead distance. Section 4 presents the traffic flow models developed for ATMA system operation. In Section 5, the data collected from field testing are introduced and, then, used to calibrate and validate the developed models. The modeling outcomes and implications to ATMA system operation are also presented. Section 6 concludes this manuscript and discusses how to deploy the designed operation guidelines.

2. LITERATURE REVIEW

ATMA is a quickly emerging technology that combines the usage of CAV and TMA in work zones. While CAV has the great potential of changing our daily life and has attracted significant research attention recently [8, 9], the question of when it can be widely used in common passenger vehicles is yet to be determined. On the other hand, the application of CAV in a narrowly defined and simplified environment, such as work zone locations with the lead-follower autonomous driving concept, becomes more realistic. ATMA has been receiving a significant amount of attention from state DOTs with the hope to reduce worker fatalities in work zones. Colorado, and Missouri are among the first States in the U.S. to test and deploy ATMA vehicles to remove DOT workers from the driver seat [7, 8, 10, 11]. Several other States, including California, Minnesota, Virginia, Ohio, Tennessee [12] are in the process of testing, developing, or deploying similar technologies. In addition, Colorado DOT is leading an autonomous maintenance technology pool fund with 13 State DOT members [13].

Many car-following models have been developed to illustrate car-following behavior. They describe how a leading vehicle and a following vehicle interact with each other, which is an important consideration to ensure a safe driving experience in a roadway network. The Gazis-Herman-Rothery (GHR) model was first formulated in 1958 at General Motors research laboratory in Detroit [14]. This model related a vehicle's acceleration to the speed of the leader

vehicle, relative speed and spacing between the follower and the leader vehicles, and driver reaction time. The collision avoidance (CA) model was firstly proposed by Kometani and Sasaki [15]. This model described the safe following distance as a quadratic function of the speeds of the follower and leader vehicles and reaction time, and the four parameters that needed to be calibrated. This model was then further improved by Gipps, in which several mitigating factors were considered [16]. The Gipps model can be calibrated using more common-sense assumptions about driver behavior when compared with the previous CA model.

A simplified car-following model was proposed by Newell, in which a follower's trajectory was a simple translation of its leader's trajectory by a specific distance and a time [17]. The relationship between spacing and velocity for a single vehicle was linear related to the specific distance and time. The Newell car-following model has been empirically validated in several studies. For example, it was verified by measuring vehicles discharging from long queues at signalized intersections [18], the jth vehicle follows the same trajectory as its leading (the j-1th) vehicle, except for a translation in time and space. Ten different car-following models, including Newell's model, were tested with data collected via Differential Global Positioning System (GPS)-equipped cars on a test track in Japan [19]. The results indicated that the validation process gives acceptable errors from 17% to 22%. In another study [20], four car-following models were calibrated and validated using trajectory data of four vehicles in a platoon under real traffic conditions on both urban and extraurban roads. The results showed that Newell's car-following model, the simplest model, performed the best in a comparison of the four models. Due to its simplicity, the Newell's car-following model has been applied widely in many studies. A parsimonious method was proposed to estimate the parameters of Newell's car-following model in congestion [21]. Analysis of the I-80 NGSIM dataset suggested that driving behaviors vary from one vehicle to the another. Newell's car-following model was also applied on simulated trajectories to quantify the individual linear speed-spacing relationships [22], and was shown to have direct linkage to the macroscopic LWR model [23, 24].

Considering those good properties, in the sections below, Newell's car-following model is adopted in this manuscript, but with some modifications. One basic assumption of the Newell's car-following model is that a vehicle following the leader vehicle in a homogenous space replicates the trajectory of the leader vehicle with a constant time and distance offset. However, many recent studies consider heterogenous driving behaviors. Several studies [25-28] have confirmed that parameters calibrated in car-following models can be different for different drivers. Focusing on Newell's car-following model, Chiabaut, Leclercq [28] examined and calibrated interdriver heterogeneity through a proposed effective estimation method. Other studies [29, 30] also found evidence which suggested that a single driver's actions can be better described using different car-following parameters and/or different models to describe a single vehicle trajectory. Focusing on Newell's car-following model, Ahnet al. [31] discussed how speed-spacing relationship can evolve during acceleration or deceleration phase. They considered varying wave speed within driver due to stochasticity, and identified wave paths through matching points with similar speed. Zheng et al. [32, 33] further examined the changes in the speed-spacing parameters under major disturbances such as lane changes and traffic flow oscillations. Furthermore, Tayloret al. [34] extended Newell's car-following model and calibrated the time-dependent parameters using the Dynamic Time Warping algorithm using NGSIM dataset. The experiment results showed a decrease in the response time and critical spacing during a deceleration period, and then recovered back to some near-steady-state condition. Thus, we assume the critical spacing is varying with driver's speed, but with a

constant response time for simplicity. This assumption is also based on Helly's linear car-following model [35], in which the desired following distance is a linear function of the vehicle velocity and acceleration.

In terms of lane-changing (LC) behavior decisions, a number of models were proposed to capture a driver's decision on whether or not to execute a lane change. For example, Gipps [36] proposed that a driver's lane changing behavior in an urban street was governed by two basic considerations: maintaining a desired speed and being in the correct lane. Gipps' model considered the driver behavior as deterministic, so that a driver decided to maintain the desired speed or be in the correct lane based on the distance to the intended turn. By extending Gipps model to freeways, Yang and Koutsopoulos [37] classified lane changing behavior as mandatory or discretionary, and modeled the process as four sequential steps: decision to consider a lane change, choice of the target lane, search for an acceptable gap, and execution of lane change. The lead gap was defined as the clear spacing between the front of the lane changer and the rear of the leader in the target lane, and the follow gap was defined as the clear spacing between the rear of the lane changer and the front of the follower in the target lane. The gap acceptance model examines the lead and follow gaps for performing a lane change in the target lane.

As reviewed above, although extensively studied, the above-mentioned car-following and lane-changing models were developed with a general passenger vehicle as their study object and, thus, cannot be directly applied to the ATMA vehicle system. This is because, the unique characteristics of the ATMA system, including its two-vehicle system design, autonomous driving capability, as well as the existence of gap distance between leader and follower trucks, all call for the modifications of classic models, before being used.

3. PRELIMINARIES

3.1 Spacing-Velocity Relationship with Newell Car-following Model

In this section, we show how to analytically derive the car-following distance for a general vehicle. As mentioned above, in this study we adopt the Newell's simplified car-following model [17], which is originally developed to describe a passenger vehicle's car-following behavior. Figure 2 depicts its characteristics, in which a leading vehicle $n-1$ initially drives at a velocity v_1 and then changes to another velocity v_2. The actual trajectory should be the dotted line, but for simplification purposes, a piece-wise linear approximation, represented by the solid line, is used. The following vehicle n travels at the velocity v_1 at first, and then changes the velocity v_2 at a turning point with a spatial delay d_n and a temporal delay τ_n. The delay d_n can be explained as the minimum distance to ensure safe driving and the τ_n can be explained as the necessary response time of driver n.

The following vehicle n's trajectory $x_n(t+\tau_n)$ can be described, as below, in which $x_{n-1}(t)$ is its leading vehicle's trajectory at time t, and τ_n and d_n represent the spatial delay and temporal delay, respectively.

$$x_n(t+\tau_n) = x_{n-1}(t) + d_n \tag{1}$$

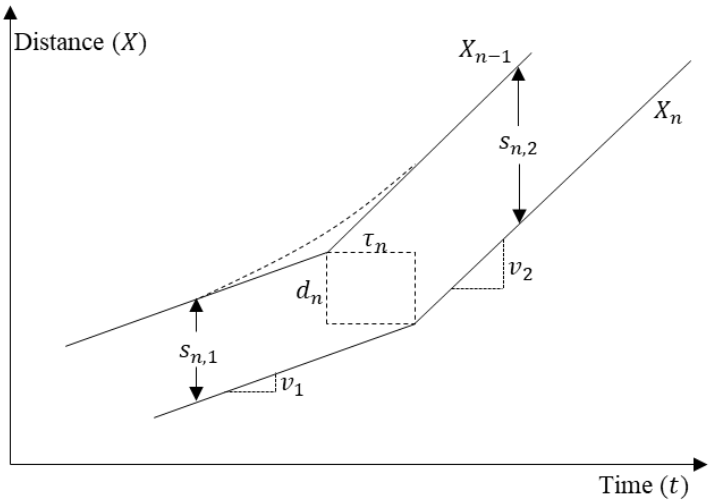


Figure 2. Time-space Diagram of Newell's Simplified Car-following Model.

Based on the Newell's car-following model, the required car-following distance for a vehicle n, at time i, can be derived as Eq. (2) below. The required car-following distance, or spacing, is denoted as $s_{n,i}$. The vehicle's speed is denoted as v_i, and d_n and τ_n are the two parameters from the Newell's car-following model.

$$s_{n,i} = d_n + v_i \cdot \tau_n \tag{2}$$

Following Newell's car-following model, Eq. (2) takes a linear form, which indicates that there exists a linear relationship between the velocity and spacing, as shown in Figure 3. When the velocity of vehicle n increases, the driver will keep a larger spacing until it reaches the free flow speed.

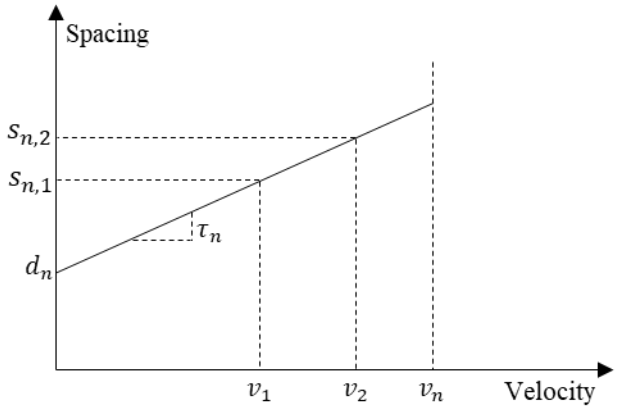


Figure 3. Relationship between Spacing and Velocity for A Single Vehicle.

3.2 Critical Gaps for Lane-changing

In this section, we show how to derive the critical gaps that are necessary for a general vehicle to switch lanes. We adopt the lane-changing models proposed by Yang and Koutsopoulos [37], which used four sequential steps to describe a passenger vehicle's lane-changing behavior decisions, including a decision to consider a lane-changing, choice of the target lane, search of an

acceptable gap, and execution of lane-changing. Once the target lane is selected, an acceptable gap is required for the driver to change lanes.

For a general vehicle to switch lanes, it needs to consider both lead gap and lag gap, as shown in Figure 4. The lead gap L_{lead} is the clear spacing between the front of the subject vehicle and the rear of the lead vehicle in the target lane 1. The lag gap L_{lag} is the clear spacing between the rear of the subject vehicle and the front of the lag vehicle in the target lane 1.

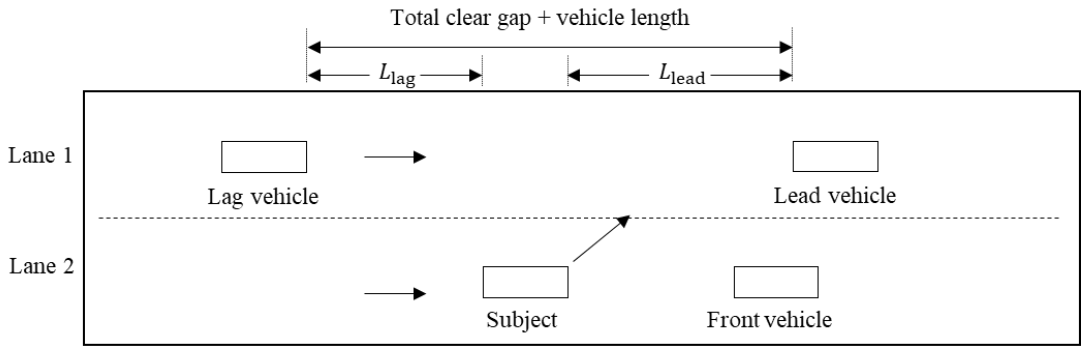


Figure 4. An Illustration of Acceptable Gap for Lane Changing.

In other words, if the subject vehicle at lane 2 intends to change lanes to the target lane 1, the available gap distance in the lane 1 needs to be higher than the summation of L_{lead}, L_{lag} and vehicle length. As such, the critical gaps L for lane changing can be expressed as Eq. (3).

$$L = L_{\text{lead}} + L_{\text{lag}} + L_{\text{vehicle length}} \tag{3}$$

3.3 Roll-ahead Distance

Roll-ahead Distance (RAD) is the distance a shadow vehicle (i.e., Truck Mounted Attenuators (TMA)) will move when struck by an errant vehicle in work zones. It is important to select a sufficient distance in advance of workers or equipment being protected. The RAD varies depending on the weights and speeds of the two vehicles involved, the extent to the extent to which the shadow vehicle is restrained, and certain pavement characteristics. All of the factors except vehicle weights and impacting vehicle speed can be accounted for with a series of assumptions. Humphreys and Sullivan gave two tables showing the expected RAD for moving and stationary operations, using the classical conservation of momentum equation [38].

In the ATMA system, the follower truck is served as a shadow vehicle to protect the DOT worker in the leader truck when performing maintenance. When the operator in the leader truck set a gap distance between the leader truck and the follower truck, the RAD should be considered. In our manuscript, we adopt the recommended shadow vehicle spacing from the American Association of State Highway and Transportation Officials (AASHTO) Roadside Design Guide [39]. According to the ATMA system configuration, the follower truck with equipment weighs 23,365 lbs. For vehicle weights in this range, AASHTO recommends that the RAD for moving operation (shown in Table 1) are 172 ft, 150 ft and 100 ft, when the speed limit are greater than 55 mph, between 45 mph and 55 mph, and less than 45 mph, respectively. Typically, Missouri DOT recommends a minimum of 150 feet RAD to the work activity during stationary operation or short duration or mobile operations [40]. Texas DOT suggests that the

minimum distance the TMA-equipped vehicle will be placed in advance of the work crew is 30 feet[41].

Table 1. Guidelines for spacing of shadow vehicles (from AASHTO Roadside Design Guide).

Weight of shadow vehicle (lb)	Operating speed (mph)	Recommended RAD for moving operation (ft)
$\geq 22{,}000$	> 55	172
	$45 - 55$	150
	< 45	100

4. MODEL DEVELOPMENT FOR ATMA SYSTEM OPERATION

In this section, we develop traffic flow models for ATMA vehicle system operation, with the goal of determining its minimum driving requirements. Specifically, technical requirements under three scenarios are investigated, including (1) the required car-following distance for a leader truck and a follower truck, respectively, while driving on a roadway segment; (2) the critical lane-changing gap distance of the system to complete a safe lane-changing action; and (3) the required intersection clearance time when crossing an intersection. The key to examination of these three scenarios is the modeling of car-following and lane-changing behavior. A highway with two lanes in each direction and with a signalized intersection is selected as the analysis target, although the model can be easily extended to other scenarios.

4.1 Minimum Car-following Distance Requirement

4.1.1 Minimum Car-following Distance for the Leader Truck

When the traffic is not congested, the operating speed of a maintenance vehicle is much slower than the other general vehicles, so for a leader truck, its car-following distance is of less concern while driving on a roadway segment. However, when the traffic is congested, or when the ATMA is approaching an intersection, the spacing between a leader truck and a general vehicle ahead becomes shorter, and the ATMA system operator needs to watch out for the minimum car-following distance.

According to Newell's simplified car-following model [17], the trajectory of the leader truck can be expressed as:

$$x_{\mathrm{lt}}(t + \tau_{\mathrm{lt}}) = x_n(t) + d_{\mathrm{lt}} \tag{4}$$

where x_{lt} and x_n are the trajectory of the leader truck and the general traffic vehicle ahead, respectively. τ_{lt} is the temporal delay, or reaction time of the driver of the leader truck, and d_{lt} denotes the spatial delay.

According to Eq. (2), the required car-following distance of a leader truck can be expressed as

$$s_{\mathrm{lt}} = v_{\mathrm{lt}} \cdot \tau_{\mathrm{lt}} + d_{\mathrm{lt}} \tag{5}$$

In Eq. (5) s_{lt} is the required car-following distance for the leader truck. Considering heterogenous driving behaviors discussed in the literature review, we assume that the spatial delay d_{lt} is varying with driver's speed. This assumption is based on the experiment results [34] which shows a decrease in the critical spacing during a deceleration period. In this manuscript, we focus on the minimum required car-following distance to maintain safety. Thus, the spatial delay is further assumed to be the distance that the leader truck traverses from the moment it initiates a brake to the complete stop. This assumption is also based on Helly's linear car-following model [35], in which the desired following distance is a linear function of the vehicle velocity and acceleration. Therefore, the minimum required car-following distance s_{lt} for leader truck consists of two terms: 1) $v_{\mathrm{lt}} \cdot \tau_{\mathrm{lt}}$ is the distance the leader truck travels before a brake is initialed, in which τ_{lt} is the necessary reaction time of the driver of the leader truck; and 2) d_{lt} is the distance that the leader truck travel from initialing a brake to a full stop. In this process, as the leader truck starts at a speed of v_{lt} and then comes to a full stop, d_{lt} can be calculated by Eq. (6).

$$d_{\mathrm{lt}} = \frac{{v_{\mathrm{lt}}}^2 - 0}{2 \cdot \alpha_{\mathrm{lt}}} = \frac{{v_{\mathrm{lt}}}^2}{2\alpha_{\mathrm{lt}}} \tag{6}$$

where α_{lt} is the deceleration of the leader truck and v_{lt} is the operation speed of the leader truck.

4.1.2 Minimum Car-following Distance for the Follower Truck

For a follower truck, due to its autonomous driving nature, the first part of Eq. (5), i.e., the distance traveled during the reaction time equals 0. Once a brake request is initiated, the computer will start to brake without any time delay. As such, when compared with the leader truck, the required minimum car-following distance only includes d_{ft}. On the other hand, what's coming into play is the ATMA system's accuracy and stability in keeping the gap distance between the leader truck and the follower truck. In order words, if the system operator specifies a command following distance, we also need to consider how accurate can the follower truck keep up with this desired distance. We use ε to denote the 95% percentile of the following distance's error in the follower truck's car-following distance, and thus the minimum car-following distance becomes $d_{\mathrm{ft}} + \varepsilon$.

In addition, as mentioned above, the minimum distance for the follower truck should also consider RAD, so that in case the follower truck is hit, it won't crash into the leader truck. As such, the minimum car-following distance of the follower truck s_{ft} becomes:

$$s_{\mathrm{ft}} = \max\left(\mathrm{RAD}, d_{\mathrm{ft}} + \varepsilon\right) \tag{7}$$

4.2 Critical Gap Distance Requirement for Lane-changing

The lane-changing decision-making process of ATMA vehicles is very different from that of general vehicles. In other words, Eq. (3) is no longer applicable, as it is designed for general vehicles. As illustrated in Figure 5, the ATMA vehicle system consists of a leader truck and a follower truck , both of which are in lane 2 and need to switch to lane 1. Such a leader-follower system design indicates that, the system operator needs to find an acceptable gap for not only the

leader truck, but also for the follower truck, so that the entire system can make the lane-changing together without being interrupted by general traffic vehicles. As such, we can modify Eq. (3) to be Eq.(8) below, to calculate the required minimum acceptable gap distance L for ATMA system.

$$L = L_{\text{lead}} + L_{\text{lt}} + L_{\text{gap}} + L_{\text{ft}} + L_{\text{lag}} \tag{8}$$

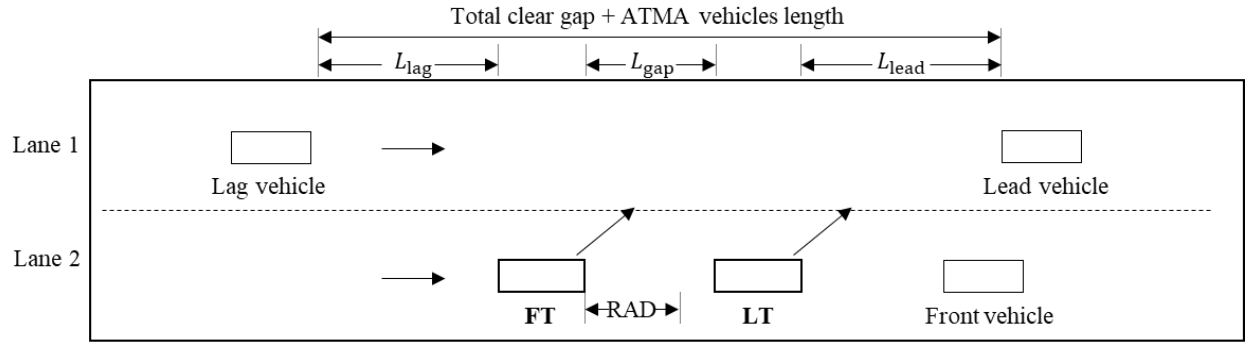


Figure 5. Critical Lane Changing Gap of ATMA Vehicles.

The minimum acceptable gap distance for the AMTA vehicles includes three components: 1) lead gap distance L_{lead}, the critical distance between the leader truck and the general vehicle in the target lane 1 (i.e., the leader vehicle), so in case an emergency happens, the leader truck can come to a safe stop; 2) critical gap distance required for the follower truck switch lanes, including ATMA vehicles' length and the gap distance between two trucks (i.e., $L_{\text{lt}} + L_{\text{gap}} + L_{\text{ft}}$); and 3) lag gap distance L_{lag}, the distance between the follower truck and the general vehicle in lane 1 (i.e., the lag vehicle), so in case an emergence happens, the lag vehicle can also come to a safe stop. To evaluate this critical lane-changing gap distance, we bring in the critical lane-changing time headway t_c. It also includes three components $t_{\text{min},1}$, $t_{\text{min},2}$, and $t_{\text{min},3}$, corresponding to the distance headway L_{lead}, $(L_{\text{lt}} + L_{\text{gap}} + L_{\text{ft}})$, and L_{lag}, respectively. Next, we show how to derive the needed time headway and gap distance.

To evaluate the lead gap distance L_{lead} and the lag gap distance L_{lag}, we firstly consider the required time headway $t_{\text{min},1}$ and $t_{\text{min},3}$. Assume, in a general case, a minimum time headway between two vehicles is t_{min}. Note due to crash avoidance requirement, the value of t_{min} relies on the driving speed v, vehicle maximum deceleration α ft/s^2, and driver's response time t_{rps}. The required time headway t_{min} can be calculated by Eq. (9).

$$t_{\text{min}} = t_{\text{rps}} + v/\alpha \tag{9}$$

As such, we can calculate required time headway for the lead gap distance and the lag gap distance, respectively. First, for a leader truck that is driving at a speed of v_{lt}, to avoid crashing into the leader vehicle in the target lane 1, the minimum time headway $t_{\text{min},1}$ is the human perception response time t_{rps} plus the necessary braking time which depends on its speed v_{lt} and its maximum deceleration value α_{lt}, as shown in Eq. (10). Then the critical lead gap distance can be estimated by Eq. (11).

$$t_{\text{min},1} = t_{\text{rps,lt}} + v_{\text{lt}}/\alpha_{\text{lt}} \tag{10}$$

$$L_{\text{lead}} = v_{\text{lt}} t_{\text{rps,lt}} + \frac{v_{\text{lt}}^2}{2\alpha_{\text{lt}}} \tag{11}$$

Second, for $t_{\min,3}$, we need to make sure the lag vehicle in lane 1 can stop in time, if necessary. We use Eq. (12) to calculate $t_{\min,3}$, in which $t_{\text{rps,gv}}$ is the perception response time of the general vehicle driver, v_{ffs} is the free flow speed, and α_{gv} is the maximum deceleration value of a general vehicle.

$$t_{\min,3} = t_{\text{rps,gv}} + v_{\text{ffs}}/\alpha_{\text{gv}} \tag{12}$$

Then the lag gap distance can be estimated by

$$L_{\text{lag}} = v_{\text{ffs}} t_{\text{rps,gv}} + \frac{v_{\text{ffs}}^2}{2\alpha_{\text{gv}}} \tag{13}$$

Finally, for $t_{\min,2}$, since the follower truck is driving autonomously and connected with the leader truck, the human perception response time is no longer needed. In other words, we only need to consider the time for follower truck to switch a lane. The system operator can set the gap distance on the user interface in the leader truck, and we denote this preset gap distance as $L_{\text{gap},0}$. However, it should be noted that the RAD should be considered in this step. The distance between a leader truck and a follower truck should always be no less than RAD. In other words, we should use $L_{\text{gap}} = \max(\text{RAD}, L_{\text{gap},0})$ to denote the distance between them. Apart from L_{gap}, ATMA vehicles length, $L_{\text{lt}} + L_{\text{ft}}$, also should be included to satisfy the critical lane changing gap shown in Figure 5. Since the ATMA driving speed v_{lt} is known, the needed time can be easily computed. $t_{\min,2}$ can then be calculated via Eq. (14).

$$t_{\min,2} = \frac{L_{\text{gap}} + L_{\text{lt}} + L_{\text{ft}}}{v_{\text{lt}}} = \frac{\max(\text{RAD}, L_{\text{gap},0}) + L_{\text{lt}} + L_{\text{ft}}}{v_{\text{lt}}} \tag{14}$$

The minimum acceptable time headway for the AMTA vehicles to safely change lanes t_c can then be calculated by Eq. (15) according to Eqs (10), (12) and (14), and minimum acceptable gap distance can be estimated by Eq. (16) according to Eqs (11) and (13). It should be noted that since leader truck and general vehicles are both driven by human drivers, the perception response time should be same, i.e., $t_{\text{rps,lt}} = t_{\text{rps,gv}}$. However, the vehicle deceleration performances between them are different, i.e., $\alpha_{\text{gv}} \neq \alpha_{\text{lt}}$.

$$t_c = 2t_{\text{rps}} + \frac{v_{\text{lt}}}{\alpha_{\text{lt}}} + \frac{\max\left(\text{RAD}, L_{\text{gap},0}\right) + L_{\text{lt}} + L_{\text{ft}}}{v_{\text{lt}}} + \frac{v_{\text{ffs}}}{\alpha_{\text{gv}}} \tag{15}$$

$$L = v_{\text{lt}} t_{\text{rps,lt}} + \frac{v_{lt}^2}{2\alpha_{\text{lt}}} + L_{\text{lt}} + \max(\text{RAD}, L_{\text{gap},0}) + L_{\text{ft}} + v_{\text{ffs}} t_{\text{rps,gv}} + \frac{v_{\text{ffs}}^2}{2\alpha_{\text{gv}}} \tag{16}$$

4.3 Intersection Clearance Time Requirement

In this subsection, we model the minimum clearance time requirement for the ATMA vehicle system at intersections, which is the amount of time needed for vehicles to safely pass an intersection. For a common general vehicle, it merely needs to follow the signal instruction to cross an intersection or to make a turn as long as it can enter the intersection before the light turns yellow, the designed intersection clearance time is usually sufficient for it go through the

intersection, or make a turn. However, for an ATMA vehicle system, this is very different. Considering the two-vehicle design, as well as the gap distance between a leader truck and a follower truck, the clearance time designed for a single general vehicle is not enough for the ATMA vehicle system. As such, the leader truck driver needs to assess whether the signal time is sufficient for the ATMA vehicles to cross the intersection.

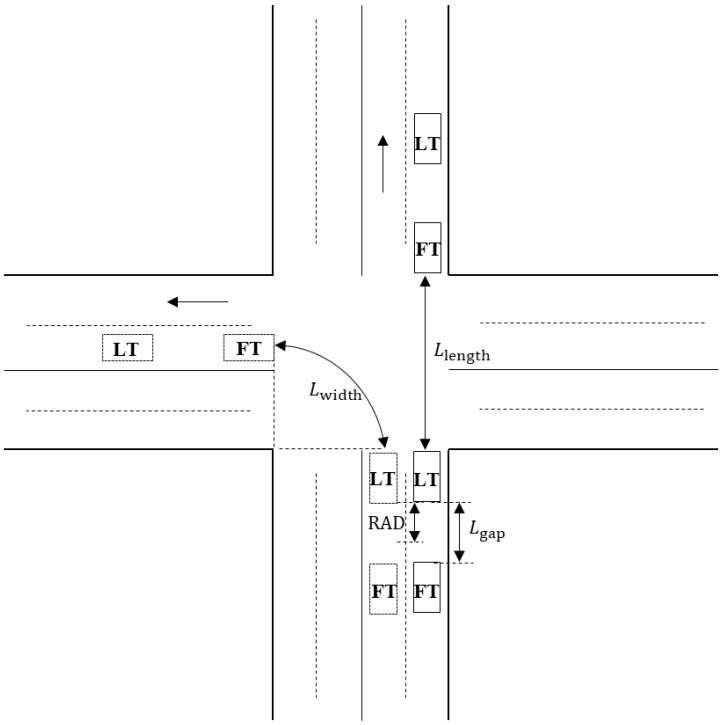


Figure 6. Four-lane Two-way Highway Intersection Profile.

The distance of crossing an intersection or making a turn is shown in Figure 6. If the ATMA vehicles need to cross the intersection, the clearance time requirement for ATMA vehicles can be calculated by

$$t_{\text{straight}} = (L_{\text{length}} + \max(\text{RAD}, L_{\text{gap},0}) + L_{\text{lt}} + L_{\text{ft}})/v_{\text{lt}} \quad (17)$$

where the L_{length} is the length of the intersection, $L_{\text{gap},0}$ is the preset gap distance between two trucks, L_{lt} and L_{ft} are the length of the leader truck and follower truck, and v_{lt} is the travel speed of ATMA vehicles.

On the other hand, if the ATMA vehicle system needs to make a left-turn at the intersection, the clearance time requirement for ATMA vehicles is calculated by

$$t_{\text{turn}} = (L_{\text{width}} + \max(\text{RAD}, L_{\text{gap},0}) + L_{\text{lt}} + L_{\text{ft}})/v_{\text{lt}} \quad (18)$$

where the L_{width} is the width of the intersection.

5. NUMERICAL ANALYSIS

In this section, we first present the data collected from field testing, which is then used to calibrate and validate the developed models. The modeling outcomes and implications to ATMA system operation are presented in the end.

5.1 Field Testing Description

A field-testing event was organized at Sedalia, Missouri in May 2019, to test the ATMA system's performance. The ATMA vehicles was provided by Missouri Department of Transportation (MoDOT), and the length of the leader truck and the follower truck are $L_{\mathrm{lt}} = L_{\mathrm{ft}} = 40$ ft. The purpose of the event was to test whether the system elements (including hardware and software components) could meet pre-defined accuracy and functional requirements for a minimum of 32 consecutive hours of operation over several days, without the need for a MoDOT operator to take control due to poor performance of the leader-follower system. The testing was performed in a controlled environment, i.e., the roadway was limited to no traffic. A total of 31 cases were planned to test the system's performance. Among them, 23 test cases were quantifiable, including 7 on follow distance and accuracy, 5 on communication loss, 3 on obstacle detection, and 8 on emergency situations. Each test was repeated for three times to assure statistical accuracy. The other 8 test cases were simple yes/no testing, such as visual inspection on the system and the trucks, data logging, turn signals, and the functionality on the user interface. The detailed testing procedure and outcomes can be found in [7, 11]. Statistical analysis results demonstrated that the ATMA system was able to function as expected, and its performance was acceptable when compared with predefined criteria.

5.2 Data Collection

Multiple hardware, including a GPS device, vehicle to vehicle (V2V) communication, LiDAR and other sensors were installed on the ATMA vehicles and, during the field testing, the vehicle status information was automatically collected and recorded. These datasets were downloaded to a working computer and processed by Python language for this research.

The leader truck log file recorded the timestamp, eCrumb message, position (latitude, longitude and altitude), heading and velocity information of the lead truck, as shown in Figure 7 (a). Whereas for the follower truck, as it was driving autonomously, its log file also recorded the autonomous driving status information, such as the actual distance between the leader truck and the follower truck, the number of GPS satellites connected, acceleration, and cross track error. Figure 7 (b) shows a screenshot of the data collected by the follower truck.

(a)

	LCB	CRUMB	STAMP	LAT	LON	ALT	HEADING	VELOCITY
40:04.1	LDR	9802	18400390	38.69313	-93.2612	-0.018	280.999	2.28
40:04.2	LDR	9803	18400400	38.69313	-93.2612	-0.015	280.959	2.42
40:04.2	LDR	9804	18400410	38.69313	-93.2612	-0.02	281.009	2.54
40:04.4	LDR	9805	18400420	38.69313	-93.2612	-0.033	280.969	2.67
40:04.4	LDR	9806	18400430	38.69313	-93.2612	-0.027	280.929	2.8
40:04.6	LDR	9807	18400440	38.69313	-93.2612	-0.018	280.929	2.93
40:04.7	LDR	9808	18400450	38.69313	-93.2612	-0.017	280.899	3.03

(b)

49:44.8	VEH	CRUMB	STAMP	LAT	LON	ALT	HEADING	HDG(Desi	VELOCITY	VEL(Desir	GAP	GAP(Desi	#SATS	VALID	CTE	ACCEL	STEER	STATE
49:45.1	FLW	0	13494510	38.69365	-93.2617	234.129	299.138	290.476	3.91	4.44	8.83	7.6	21	1	0	-100	0	IDLE
49:45.2	FLW	0	13494520	38.69365	-93.2617	234.138	298.638	290.476	3.95	4.44	8.83	7.6	21	1	0	-100	0	IDLE
49:45.3	FLW	0	13494530	38.69365	-93.2617	234.147	298.298	290.476	3.95	4.44	8.89	7.6	17	1	0	-100	0	IDLE
49:45.4	FLW	0	13494540	38.69365	-93.2617	234.143	297.808	290.476	4.03	4.44	8.92	7.6	17	1	0	-100	0	IDLE
49:45.5	FLW	0	13494550	38.69365	-93.2617	234.151	297.498	290.476	4.04	4.44	8.95	7.6	17	1	0	-100	0	IDLE
49:45.6	FLW	0	13494560	38.69365	-93.2617	234.154	297.118	290.476	4.04	4.44	8.97	7.6	17	1	0	-100	0	IDLE
49:45.7	FLW	0	13494570	38.69365	-93.2617	234.161	296.898	290.476	3.96	4.44	8.99	7.6	17	1	0	-100	0	IDLE
49:45.8	FLW	0	13494580	38.69365	-93.2617	234.165	296.438	290.476	3.87	4.44	9.03	7.6	17	1	0	-100	0	IDLE
49:45.9	FLW	0	13494590	38.69366	-93.2617	234.18	296.048	290.476	3.79	4.44	9.06	7.6	17	1	0	-100	0	IDLE

Figure 7. Screenshot of ATMA Vehicles' Log Files. (a) Lead Truck, (b) Follower Truck.

To derive the minimum required car-following distance and critical lane changing gap distance, the maximum deceleration of the leader truck and the follower truck shall be calibrated. During the field testing, emergency stop testing was performed for three times, which allowed data collection for this purpose. A technician in the leader truck pushed the emergency stop button, the ATMA vehicle initiated an emergency stop, and the stop time and distance were recorded (shown in Table 2). The speed of ATMA vehicles was set to be 10 mph and 15 mph.

When the speed was set as 10 mph, the standard deviation of the stop time was 0.09, whereas the standard deviation of the stop distance was 3.68. When the speed was set to be 15 mph, these values increased to 0.14 and 4.27, respectively. These numbers suggested that the error of the recorded stop distance was larger than that of the stop time. The accuracy of the stop time was considerably higher than that of the stop distance. Because of this, the data of stop time was used to estimate the maximum deceleration. The data analysis showed the maximum deceleration $\alpha_{\mathrm{lt}} = 12.4\ \mathrm{ft/s^2}$, which we adopt in this manuscript to estimate the minimum car-following distance and the critical lane-changing gap distance.

Table 2. The Stop Time and Distance in Emergency Stop Test.

Set GAP	**Speed**	**Stop Time**			**Standard Deviation**	**Stop Distance**			**Standard Deviation**
		Run 1	Run 2	Run 3		Run 1	Run 2	Run 3	
≥ 100′	10 mph	1.56 s	1.56 s	1.75 s	0.09	11.5′	15.75′	20.5′	3.68
	15 mph	1.91 s	2.13 s	1.78 s	0.14	31.08′	37′	26.58′	4.27

5.3 Analysis of Car Following Distance Requirement

5.3.1 Car Following Distance Requirement for Leader Truck

We first calibrate the values of the parameters that are needed. According to Eq. (5), the minimum required car-following distance of a leader truck is related to its travel speed and

temporal delay. AASHTO recommends a design criterion of 2.5 seconds for brake reaction time, which exceeds the 90th percentile of reaction time for all drivers [42], as such we set $\tau_{\mathrm{lt}} = 2.5s$. Then the minimum car-following distance of a leader truck can be derived as $s_{\mathrm{lt}} = \frac{{v_{\mathrm{lt}}}^2}{24.8} + 2.5v_{\mathrm{lt}}$ based on Eq. (5).

Based on these parameters, the minimum car-following distance of the leader truck s_{lt} is denoted by red line, shown in Figure 8. In the field test, the ATMA vehicles' operating speed is usually set to be 5 mph to 15 mph to make safety maintenance feasible. To make consistently, we set the operating speed of ATMA vehicles ranging from 5 mph to 15 mph in this manuscript. In Figure 8, X-axis is the operating speed of ATMA, and Y-axis is the minimum car-following distance responding to the operating speed. It can be found that when the operating speed of ATMA increases from 5mph to 15mph, the minimum car-following distance of the leader truck increases from nearly 20 ft to 75 ft. To ensure a safe driving experience, the minimum car-following distance should be set as 75 ft for the leader truck.

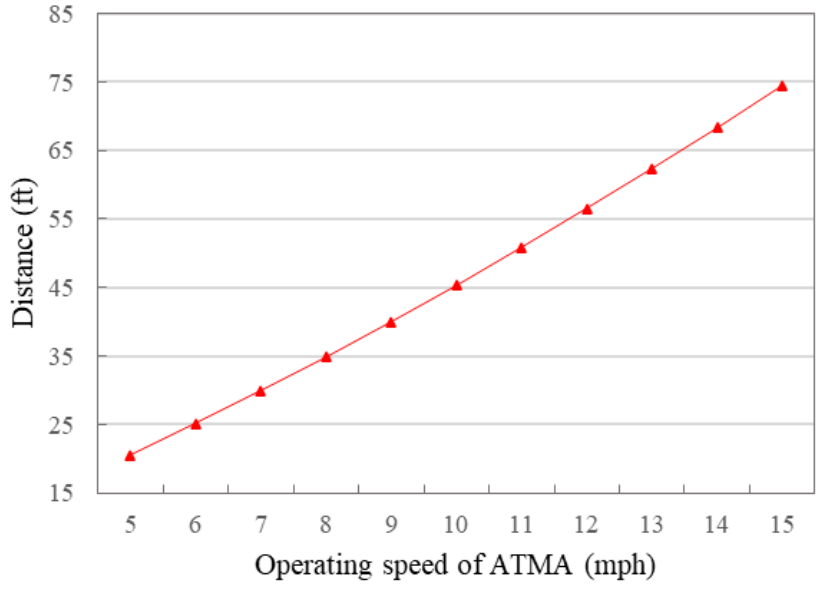


Figure 8. Minimum Car-following Distance of Leader Truck.

5.3.2 Car Following Distance Requirement for Follower Truck

For the car-following distance of the follower truck, we first calibrate the error in following distance from the field testing. The distribution of errors between the desired gap and the actual gap is plotted in Figure 9. A positive error means the actual gap is less than the desired gap, which should be included in the minimum car-following distance to ensure safe driving. The 95 percentiles of errors are found to be 6 ft, i.e., $\varepsilon = 6ft$. The minimum car-following distance of the follower truck can is thus $s_{\mathrm{ft}} = \max\left(\mathrm{RAD}, \frac{{v_{\mathrm{ft}}}^2}{24.8} + 6\right)$ according to Eq. (7).

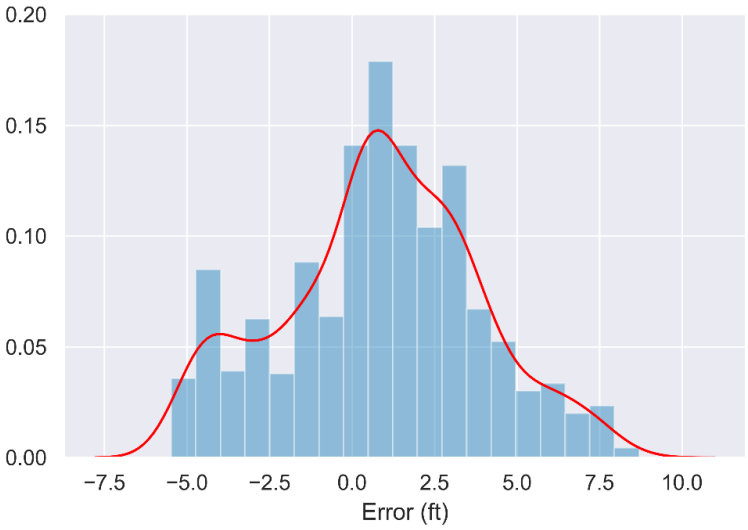


Figure 9. Frequency Distribution Histogram of Error in Follow Distance.

Based on this relationship, for the follower truck, the required minimum car-following distance s_{ft} is dominated by the roll-ahead distance. This is because for a typical ATMA cruising speed between 8mph and 15mph, the second component of Eq. (7) $\frac{{v_{\text{ft}}}^2}{24.8} + 6$ is somewhere between 6ft and 8ft, which is significantly lower than the roll-ahead distance of 100~172ft. As a result, based on Table 1 the required minimum car-following distance for the follower truck s_{ft} should be 100ft if the speed limit is lower than 45mph, or 150ft if the speed limit is between 45mph and 55mph, or 172ft if the speed limit is higher than 55mph.

5.4 Analysis of Lane Changing Critical Gap Distance Requirement

As discussed in Section 4, the critical gap distance for the ATMA vehicles includes three components: lead gap distance L_{lead} for the leader truck responding to time headway $t_{\text{min},1}$, critical gap distance ($L_{\text{lt}} + L_{\text{gap}} + L_{\text{ft}}$) required for the follower truck switch lanes responding to time headway $t_{\text{min},2}$, and lag gap distance L_{lag} for the lagging general vehicle on the target lane responding to time headway $t_{\text{min},3}$.

According to Eq.(10), the time headway $t_{\text{min},1}$ for the leader truck is related with the operation speed, maximum deceleration, and response time. According to the data from the emergency stop test, the maximum deceleration is 12.4 ft/s^2. AASHTO recommends a design criterion of 2.5 s for brake reaction time which exceeds the 90th percentile of reaction time for all drivers [42]. As such, $t_{\text{min},1}$ can be calculated by $\text{t}_{\text{min},1} = 2.5 + \frac{v_{\text{lt}}}{12.4}$. Then the critical lead gap distance can be calculated by $L_{\text{lead}} = 2.5v_{\text{lt}} + \frac{v_{\text{lt}}^2}{2\times 12.4}$. The second component, $t_{\text{min},2}$ for a follower truck, is related with the gap distance and the operation speed of ATMA vehicles as illustrated in Eq. (14) .In the ATMA system, the gap distance between the leader truck and the follower truck can be adjusted on a User Interface and ranges between 25' and 1500'. In the field testing this value was set to be 100 ft to 200 ft, which are the most common in real-world operation and thus we follow such setting to calculate the required time for a follower truck to switch lanes, per Eq. (14). As such, we have $\text{t}_{\text{min},2} = (\max(\text{RAD}, L_{\text{gap},0}) + 2 \times 40)/v_{\text{lt}}$, in which $L_{\text{gap},0}$ takes the value of 100 ft or 200 ft. The follower truck RAD takes the value of 100 ft,

150 ft and 172 ft responding to different speed limit. The third component, $t_{\min,3}$ for the lagging general vehicle on the target lane, can be derived with free flow speed, maximum deceleration and response time of the general vehicles, as illustrated in Eq. (12). Generally, the free flow speed ranges between 35 mph and 70 mph. Research suggests that approximately 90 percent of all drivers decelerate at rates greater than $11.2\ \mathrm{ft/s^2}$ (a comfortable deceleration for most driver), which is recommended as the threshold to determine stopping sight distance [42]. Studies documented in the literature [43] show that most drivers decelerate at a rate less than $14.8\ \mathrm{ft/s^2}$ when confronted with the need to stop for an unexpected object in the roadway. As such we take $14.8\ \mathrm{ft/s^2}$ as the threshold to use. $t_{\min,3}$ now becomes $t_{\min,3} = 2.5 + \frac{v_{\mathrm{ffs}}}{14.8}$. The critical lag gap distance can be derived by $L_{\mathrm{lag}} = 2.5v_{\mathrm{ffs}} + \frac{v_{\mathrm{ffs}}^2}{2\times 14.8}$ according to Eq. (13).

In total, the critical time headway for the ATMA vehicle to safely change lanes can be calculated by $t_c = 5 + \frac{v_{\mathrm{lt}}}{12.4} + \frac{\max(\mathrm{RAD}, L_{\mathrm{gap},0}) + 2\times 40}{v_{\mathrm{lt}}} + \frac{v_{\mathrm{ffs}}}{14.8}$ according to Eq. (15). Thus, the critical gap distance for lane changing can be derived by $L = 2.5v_{\mathrm{lt}} + \frac{v_{\mathrm{lt}}^2}{2\times 12.4} + \max(\mathrm{RAD}, L_{\mathrm{gap},0}) + 2\times 40 + 2.5v_{\mathrm{ffs}} + \frac{v_{\mathrm{ffs}}^2}{2\times 14.8}$ according to Eq. (16).

The resulted lane-changing critical time headway and gap distance are illustrated in Figure 10, with the distance between a leader truck and a follower truck set to be 100 ft shown in Figure 10 (a) and 200 ft shown in Figure 10 (b). The free flow speed is set to be 35 mph, 50 mph and 70 mph under different scenarios, thus the follower truck RAD takes values of 100 ft, 150 ft and 172 ft, respectively. When the gap distance between two trucks is set to be 100 ft, as illustrated in Figure 10 (a)-1, the required critical time headway reduces when ATMA operation speed increases, which mainly caused by the time headway for gap distance between ATMA vehicles decreasing. While the required critical gap distance slightly increases when ATMA operation speed increases, which caused by the lead gap distance increasing shown in Figure 10 (a)-2. It also can be observed that a higher free flow speed requires a higher critical time headway and gap distance. Similar pattern can be found in the Figure 10 (b)-1 and (b)-2 when the gap distance between two trucks is set to be 200 ft.

For a typical freeway with 70 mph FFS, when the gap distance between two trucks is set to be 100 ft, the critical time headway ranges from 25 to 47 seconds and the critical gap distance ranges from 886 ft to 940 ft. If the gap distance is set to be 200 ft shown in Figure 10 (b)-1 and (b)-2, the critical time headway gap ranges from 26 to 51 seconds and the critical gap distance range from 914 ft to 968 ft. If we set the ATMA operation speed as 10 mph, which is the most commonly-seen scenario, the required critical time headway becomes 30 seconds and 32 seconds in subfigures (a)-1 and (b)-1, respectively. Moreover, the required critical gap distance becomes 912 ft and 939 ft in subfigures (a)-2 and (b)-2, respectively. When compared with a common passenger vehicle, these numbers are significantly higher which, again, confirm our previous hypotheses that an ATMA system operator needs to drive a vehicle in a very different way than when driving a common vehicle. Supplemental work zone traffic management actions, such as traffic cones or flaggers, might be helpful to ensure this lane-changing action not be interrupted by other vehicles.

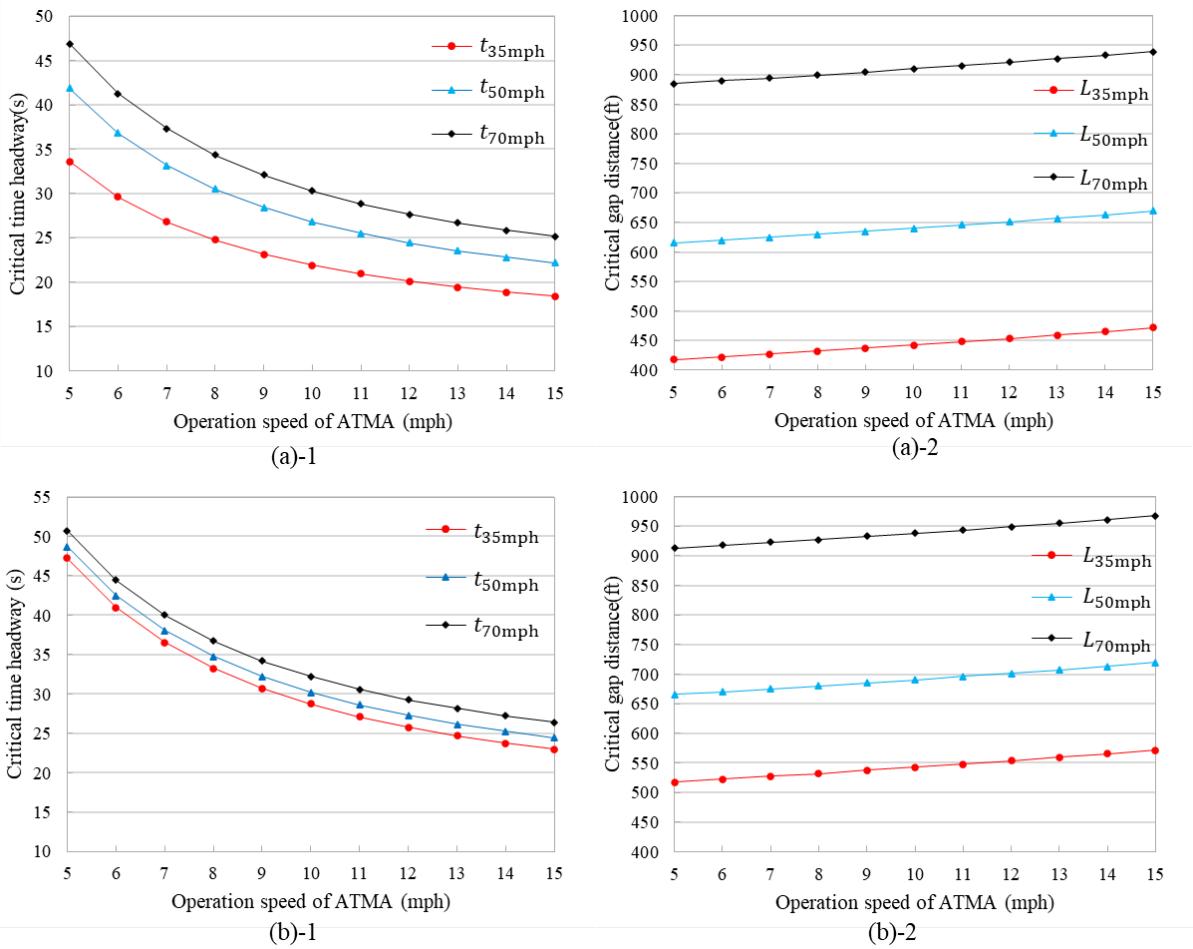


Figure 10. Critical Time Headway and Gap Distance Requirement for Lane Changing Subject to RAD Requirement: (a) 100 ft Gap Distance; (b) 200 ft Gap Distance.

5.5 Analysis of Intersection Clearance Time Requirement

We continue to use the same roadway configuration that is specified in Figure 6. The lane width is set to be 12ft so that the intersection length is 48ft, and the intersection width is $\pi \times (24 + 6) \times 2/4 = 47.1$ ft. The gap distance between leader truck and follower truck is also set to be 100 ft and 200 ft. The required time for ATMA vehicles to cross the intersection can be calculated by $t_{\text{straight}} = (48 + \max(\text{RAD}, L_{\text{gap},0}) + 2 \times 40)/v_{\text{lt}}$ based on Eq.(17). Similarly, the required time for ATMA vehicles to make a left turn is calculated by $t_{\text{turn}} = (47.1 + \max(\text{RAD}, L_{\text{gap},0}) + 2 \times 40)/v_{\text{lt}}$ based on Eq. (18).

Figure 11 (a) and (b) present the required time for ATMA to pass the intersection. As the intersection length is about the same with the width, the required time for crossing intersection or making a left turn are almost equal. We can find that the required time to cross this intersection ranges from 10 to 31 seconds when the gap distance is 100 ft, or 13~38 seconds with gap distance of 150 ft, or 15~45 seconds with gap distance of 200 ft. In particular, if the ATMA operation speed is set to 10 mph which is the most commonly-seen, the required intersection clearance time ranges from 16 seconds to 22 seconds for through-movement and a left-turn.

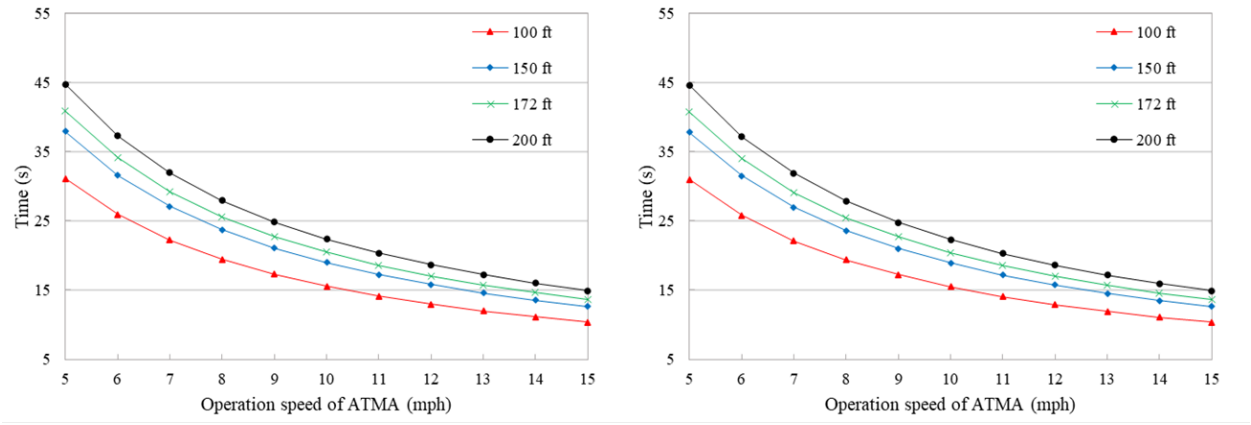


Figure 11. Required Time for ATMA Vehicles at An Intersection to: (a) go straight; (b) turn left.

6. CONCLUSION AND DISCUSSION

This manuscript focuses on modeling and developing a set of rules and instructions to operate the ATMA vehicle system, particularly when it comes to critical locations where correct decision making is needed. Different from general vehicles, the ATMA vehicle system consists of a follower truck and a leader truck, with some distance between them. The operators are required to make driving decisions, not only from the leader truck's perspective, but they must also consider the potential implications of their decisions on the follower truck. Specifically, three technical requirements are investigated, including those for car-following distance, critical lane-changing gap distance, and intersection clearance time. The Newell car-following model and the classic lane-changing behavior model are modified to model the driving behaviors of the ATMA vehicles at those critical decision-making locations. Data are collected from real-world field testing to calibrate and validate the developed models.

The modelling outputs suggest important thresholds for ATMA system operators to follow, in order to ensure the safe driving of both public and ATMA vehicles. The results suggested a minimum car-following distance of 75 ft for a leader truck. The minimum car-following distance for a follower truck is dominated by roll-ahead distance and ranges from 100ft to 172ft depending on the speed limit. For a typical freeway with free flow speed 70 mph and ATMA operating speed 10 mph, the system requires a minimum gap distance of 912 ft and 939 ft in the target lane to perform a safe lane-changing when the distance between a leader truck and a follower truck is 100 ft and 200 ft, respectively. With regard to the intersection clearance time, when the gap distance between a leader truck and a follower truck is set to be 100 ft, the system requires 15 seconds to cross an intersection or to safely make a right turn, and this number increases to 25 seconds if the distance between a leader truck and a follower truck increases to 200 ft. When compared with a common passenger vehicle, these numbers are significantly higher, which highlights the importance of using the modeling outcomes to train ATMA system operators, as well as to provide supplemental work zone traffic management actions to work with the operation of ATMA vehicles to ensure a safe and smooth operation.

The designed guidelines are expected to be deployed through combing with CAV technology, which has been widely adopted in practice. ATMA vehicles, as an application of CAV technology, are installed with many sensors, e.g., Radar, Lidar, Ultrasonic and Camera, etc.,

to detect the obstacle and the distance. Also, the ATMA vehicles can also apply the Vehicle-to-Infrastructure (V2I) communication, which is the wireless exchange of data between vehicles and road infrastructure. The sensors, installed at the front of and both sides of trucks, can detect different gap distance as discussed in Chapter 4. Those detected gap distance can be displayed on the User Interface in the leader truck, so that the driver can determine whether the realistic gap distance satisfies the minimum car-following distance or critical gap for lane changing. At the intersection, the leader truck can receive the information from the traffic signal controller by V2I communication. When the traffic light is green, the driver can make a decision once knowing how much green time left and whether it is enough for ATMA vehicles go through or make a left turn.

This study might be improved in the following aspects. First, the car-following behavior in reality may not strictly follow the Newell car-following model. More accurate calibration of the driving behavior might be helpful. Second, the field testing is performed with the ATMA system provided by Micro Systems Inc. (MSI), a wholly owned subsidiary of Kratos Defense and Security Solutions (Kratos-MSI). The vehicle performance, especially the deceleration performances and system accuracy in keep car-following distance might be different, if a product from a different company is used.

7. ACKNOWLEDGEMENT

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8. AUTHOR CONTRIBUTIONS

The authors confirm contribution to the paper as follows: study conception and design: Q. Tang, X. Hu, R. Qin; data collection: Q. Tang, X. Hu; analysis and interpretation of results: Q. Tang; draft manuscript preparation: Q. Tang, X. Hu. Manuscript review and editing: X. Hu, R. Qin. All authors reviewed the results and approved the final version of the manuscript.

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